MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT PUMP SHED

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2016

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	31-07-2020	31-08-2020	26-10-2023	Date	31-07-2020	31-08-2020	26-10-2023
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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C01

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Column Foundation for **PUMP SHED.**

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for PUMP SHED
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008
Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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3 Design of Pump Shed Column Foundation

3.1 Design of Foundation mkd. F1

Load Detail	Col Mkd. B3		
	FX (T)	FY (T)	FZ (T)
1 DL	1.0	3.2	0.0
2 LLROOF	0.4	1.6	0.0
30 MR	0.0	3.0	0.0
21 WL1(+X)CASE-A	0.9	0.6	0.0
22 WL2(+X)CASE-B	0.6	-0.9	0.0
23 WL3(-X)CASE-A	-0.6	0.7	0.0
24 WL4(-X)CASE-B	-0.9	-0.7	0.0
25 WL5(+Z)CASE-A	-0.6	-2.6	0.0
26 WL6(+Z)CASE-B	-0.7	-2.8	0.0
27 WL7(-Z)CASE-A	-0.6	-2.6	0.0
28 WL8(-Z)CASE-B	-0.7	-2.8	0.0

(Foundation has been designed for the most critical support reactions)

(Refer Foundation Detail dwg no. MD-216-49PR-EG-CV-GAD-2017 & Foundation Load Data)

Strength Load Combinations as per Cl. No. 2.3.2, SBC 301:2007 for the most critical support reactions as follows:

STRENGTH DESIGN	Col Mkd. B3		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	1.36	4.49	0.00
1.4D+1.7(MR)+0.5Lr	1.6	10.4	0.0
1.2D+1.6Lr+1.0(MR)	1.8	9.4	0.0
1.2D + 1.6WL1+0.5Lr	2.8	5.6	0.0
1.2D + 1.6WL2+0.5Lr	2.3	3.3	0.0
1.2D + 1.6WL3+0.5Lr	0.4	5.8	0.0
1.2D + 1.6WL4+0.5Lr	-0.1	3.5	0.0
1.2D + 1.6WL5+0.5Lr	0.4	0.5	0.0
1.2D + 1.6WL6+0.5Lr	0.3	0.2	0.0
1.2D + 1.6WL7+0.5Lr	0.4	0.5	0.0
1.2D + 1.6WL8+0.5Lr	0.3	0.2	0.0
0.9D+1.6WL1	2.3	3.9	0.0
0.9D+1.6WL2	1.8	1.5	0.0
0.9D+1.6WL3	0.0	4.0	0.0
0.9D+1.6WL4	-0.6	1.7	0.0
0.9D+1.6WL5	-0.1	-1.3	0.0
0.9D+1.6WL6	-0.2	-1.6	0.0
0.9D+1.6WL7	-0.1	-1.3	0.0
0.9D+1.6WL8	-0.2	-1.6	0.0

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DESIGN OF FOUNDATION MKD. F1

	ABOUT-X	ABOUT-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

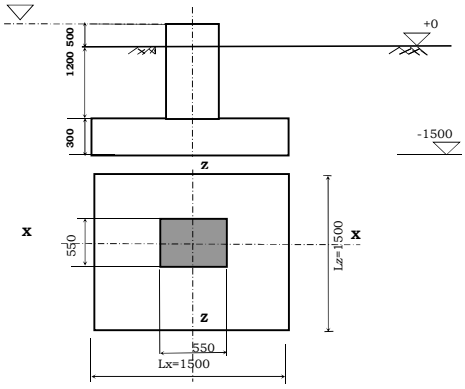
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Monorail Load	MR

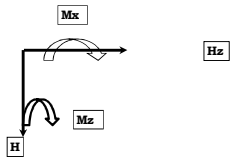
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.5*1.5*0.3*2.5)+(0.55*0.55*1.7*2.5) = 2.97T

Weight of soil = ((1.5*1.5)-(0.55*0.55))*1.2*1.8 = 4.21T

U/S Base Pl Total = 7.18T





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 217mm = 0.217m

Effective depth of slab along Z direction = 201mm = 0.201m

Clear projection along Z direction = 0.475m

Clear projection along X direction = 0.475m

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	1.36	4.83	0.00	0.0	0.00	12.01	1.36	0.00	0.00	2.72
DL+MR	0.97	6.21	0.00	0.0	0.00	13.39	0.97	0.00	0.00	1.94
DL+0.75*MR+0.75*Lr	1.26	6.68	0.00	0.0	0.00	13.86	1.26	0.00	0.00	2.53
DL+0.75*Lr+0.75*WL1	1.95	4.88	0.00	0.0	0.00	12.06	1.95	0.00	0.00	3.89
DL+0.75*Lr+0.75*WL2	1.68	3.79	0.00	0.0	0.00	10.97	1.68	0.00	0.00	3.37
DL+0.75*Lr+0.75*WL3	0.84	4.94	0.00	0.0	0.00	12.12	0.84	0.00	0.00	1.67
DL+0.75*Lr+0.75*WL4	0.57	3.87	0.00	0.0	0.00	11.05	0.57	0.00	0.00	1.15
DL+0.75*Lr+0.75*WL5	0.79	2.47	0.00	0.0	0.00	9.65	0.79	0.00	0.00	1.58
DL+0.75*Lr+0.75*WL6	0.75	2.32	0.00	0.0	0.00	9.50	0.75	0.00	0.00	1.51
DL+0.75*Lr+0.75*WL7	0.79	2.47	0.00	0.0	0.00	9.65	0.79	0.00	0.00	1.58
DL+0.75*Lr+0.75*WL8	0.75	2.32	0.00	0.0	0.00	9.50	0.75	0.00	0.00	1.51
0.6*DL+WL1	1.49	2.54	0.00	0.0	0.00	9.72	1.49	0.00	0.00	2.98
0.6*DL+WL2	1.14	1.08	0.00	0.0	0.00	8.26	1.14	0.00	0.00	2.28
0.6*DL+WL3	0.01	2.61	0.00	0.0	0.00	9.79	0.01	0.00	0.00	0.02
0.6*DL+WL4	-0.34	1.19	0.00	0.0	0.00	8.37	-0.34	0.00	0.00	0.68
0.6*DL+WL5	-0.05	-0.68	0.00	0.0	0.00	6.50	-0.05	0.00	0.00	0.10
0.6*DL+WL6	-0.10	-0.88	0.00	0.0	0.00	6.30	-0.10	0.00	0.00	0.20
0.6*DL+WL7	-0.05	-0.68	0.00	0.0	0.00	6.50	-0.05	0.00	0.00	0.10
0.6*DL+WL8	-0.10	-0.88	0.00	0.0	0.00	6.30	-0.10	0.00	0.00	0.20

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	12.01	1.36	2.72	0.00	0.00	3.31	NA	3.53	NA
DL+MR	13.39	0.97	1.94	0.00	0.00	5.18	NA	5.52	NA
DL+0.75*MR+0.75*Lr	13.86	1.26	2.53	0.00	0.00	4.12	NA	4.39	NA
DL+0.75*Lr+0.75*WL1	12.06	1.95	3.89	0.00	0.00	2.33	NA	2.48	NA
DL+0.75*Lr+0.75*WL2	10.97	1.68	3.37	0.00	0.00	2.44	NA	2.61	NA
DL+0.75*Lr+0.75*WL3	12.12	0.84	1.67	0.00	0.00	5.44	NA	5.80	NA
DL+0.75*Lr+0.75*WL4	11.05	0.57	1.15	0.00	0.00	7.24	NA	7.72	NA
DL+0.75*Lr+0.75*WL5	9.65	0.79	1.58	0.00	0.00	4.58	NA	4.88	NA
DL+0.75*Lr+0.75*WL6	9.50	0.75	1.51	0.00	0.00	4.73	NA	5.05	NA
DL+0.75*Lr+0.75*WL7	9.65	0.79	1.58	0.00	0.00	4.58	NA	4.88	NA
DL+0.75*Lr+0.75*WL8	9.50	0.75	1.51	0.00	0.00	4.73	NA	5.05	NA
0.6*DL+WL1	9.72	1.49	2.98	0.00	0.00	2.44	NA	2.60	NA
0.6*DL+WL2	8.26	1.14	2.28	0.00	0.00	2.71	NA	2.89	NA
0.6*DL+WL3	9.79	0.01	0.02	0.00	0.00	305.81	NA	326.20	NA
0.6*DL+WL4	8.37	-0.34	0.68	0.00	0.00	9.28	NA	9.90	NA
0.6*DL+WL5	6.50	-0.05	0.10	0.00	0.00	50.75	NA	54.13	NA
0.6*DL+WL6	6.30	-0.10	0.20	0.00	0.00	24.09	NA	25.70	NA
0.6*DL+WL7	6.50	-0.05	0.10	0.00	0.00	50.75	NA	54.13	NA
0.6*DL+WL8	6.30	-0.10	0.20	0.00	0.00	24.09	NA	25.70	NA

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ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	1.36	4.49	0.00	0.0	0.00	11.674	0.00	2.72
1.4D+1.7(MR)+0.5Lr	1.55	10.40	0.00	0.0	0.00	17.584	0.00	3.11
1.2D+1.6Lr+1.0(MR)	1.79	9.44	0.00	0.0	0.00	16.624	0.00	3.58
1.2D + 1.6WL1+0.5Lr	2.82	5.64	0.00	0.0	0.00	12.818	0.00	5.63
1.2D + 1.6WL2+0.5Lr	2.26	3.30	0.00	0.0	0.00	10.482	0.00	4.51
1.2D + 1.6WL3+0.5Lr	0.45	5.75	0.00	0.0	0.00	12.930	0.00	0.89
1.2D + 1.6WL4+0.5Lr	-0.11	3.48	0.00	0.0	0.00	10.658	0.00	0.23
1.2D + 1.6WL5+0.5Lr	0.35	0.49	0.00	0.0	0.00	7.666	0.00	0.70
1.2D + 1.6WL6+0.5Lr	0.27	0.17	0.00	0.0	0.00	7.346	0.00	0.54
1.2D + 1.6WL7+0.5Lr	0.35	0.49	0.00	0.0	0.00	7.666	0.00	0.70
1.2D + 1.6WL8+0.5Lr	0.27	0.17	0.00	0.0	0.00	7.346	0.00	0.54
0.9D+1.6WL1	2.33	3.87	0.00	0.0	0.00	11.045	0.00	4.66
0.9D+1.6WL2	1.77	1.53	0.00	0.0	0.00	8.709	0.00	3.54
0.9D+1.6WL3	-0.04	3.98	0.00	0.0	0.00	11.157	0.00	0.08
0.9D+1.6WL4	-0.60	1.71	0.00	0.0	0.00	8.885	0.00	1.20
0.9D+1.6WL5	-0.14	-1.29	0.00	0.0	0.00	5.893	0.00	0.27
0.9D+1.6WL6	-0.22	-1.61	0.00	0.0	0.00	5.573	0.00	0.43
0.9D+1.6WL7	-0.14	-1.29	0.00	0.0	0.00	5.893	0.00	0.27
0.9D+1.6WL8	-0.22	-1.61	0.00	0.0	0.00	5.573	0.00	0.43

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Check for Bearing Pressure

Load Combination & Calculation of Base pressure

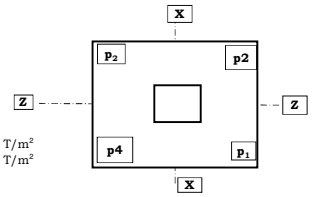
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² (BH28) For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m²) V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	Remarks
DL+Lr	12.01	2.72	0.00	5.34	4.84	0.00	12.17	12.17	2.50	2.50	9.47	O.K
DL+MR	13.39	1.94	0.00	5.95	3.45	0.00	11.40	11.40	4.50	4.50	8.70	O.K
DL+0.75*MR+0.75*Lr	13.86	2.53	0.00	6.16	4.49	0.00	12.65	12.65	3.67	3.67	9.95	O.K
DL+0.75*Lr+0.75*WL1	12.06	3.89	0.00	5.36	6.92	0.00	14.28	14.28	0.45	0.45	11.58	O.K
DL+0.75*Lr+0.75*WL2	10.97	3.37	0.00	4.87	5.98	0.00	12.86	12.86	0.89	0.89	10.16	O.K
DL+0.75*Lr+0.75*WL3	12.12	1.67	0.00	5.38	2.97	0.00	10.35	10.35	4.42	4.42	7.65	O.K
DL+0.75*Lr+0.75*WL4	11.05	1.15	0.00	4.91	2.04	0.00	8.95	8.95	4.88	4.88	6.25	O.K
DL+0.75*Lr+0.75*WL5	9.65	1.58	0.00	4.29	2.81	0.00	9.10	9.10	3.48	3.48	6.40	O.K
DL+0.75*Lr+0.75*WL6	9.50	1.51	0.00	4.22	2.68	0.00	8.90	8.90	3.55	3.55	6.20	O.K
DL+0.75*Lr+0.75*WL7	9.65	1.58	0.00	4.29	2.81	0.00	9.10	9.10	3.48	3.48	6.40	O.K
DL+0.75*Lr+0.75*WL8	9.50	1.51	0.00	4.22	2.68	0.00	8.90	8.90	3.55	3.55	6.20	O.K
0.6*DL+WL1	9.72	2.98	0.00	4.32	5.30	0.00	11.62	11.62	1.01	1.01	8.92	O.K
0.6*DL+WL2	8.26	2.28	0.00	3.67	4.06	0.00	9.73	9.73	1.61	1.61	7.03	O.K
0.6*DL+WL3	9.79	0.02	0.00	4.35	0.04	0.00	6.39	6.39	6.31	6.31	3.69	O.K
0.6*DL+WL4	8.37	0.68	0.00	3.72	1.20	0.00	6.92	6.92	4.52	4.52	4.22	O.K
0.6*DL+WL5	6.50	0.10	0.00	2.89	0.17	0.00	5.06	5.06	4.72	4.72	2.36	O.K
0.6*DL+WL6	6.30	0.20	0.00	2.80	0.35	0.00	5.15	5.15	4.45	4.45	2.45	O.K
0.6*DL+WL7	6.50	0.10	0.00	2.89	0.17	0.00	5.06	5.06	4.72	4.72	2.36	O.K
0.6*DL+WL8	6.30	0.20	0.00	2.80	0.35	0.00	5.15	5.15	4.45	4.45	2.45	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

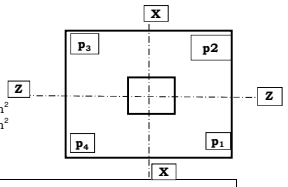
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)
1.4D	11.7	2.72	0.00	5.19	4.83	0.00	12.02	12.02	2.36	2.36	9.32
1.4D+1.7(MR)+0.5Lr	17.6	3.11	0.00	7.82	5.52	0.00	15.34	13.34	2.29	2.29	13.18
1.2D+1.6Lr+1.0(MR)	16.6	3.58	0.00	7.39	6.36	0.00	15.75	13.75	1.03	1.03	13.59
1.2D + 1.6WL1+0.5Lr	12.8	5.63	0.00	5.70	10.01	0.00	17.71	15.71	-4.31	-4.31	15.55
1.2D + 1.6WL2+0.5Lr	10.5	4.51	0.00	4.66	8.02	0.00	14.68	12.68	-3.36	-3.36	12.52
1.2D + 1.6WL3+0.5Lr	12.9	0.89	0.00	5.75	1.59	0.00	9.34	7.34	4.16	4.16	7.18
1.2D + 1.6WL4+0.5Lr	10.7	0.23	0.00	4.74	0.40	0.00	7.14	5.14	4.34	4.34	4.98
1.2D + 1.6WL5+0.5Lr	7.7	0.70	0.00	3.41	1.25	0.00	6.66	4.66	2.16	2.16	4.50
1.2D + 1.6WL6+0.5Lr	7.3	0.54	0.00	3.26	0.96	0.00	6.23	4.23	2.30	2.30	4.07
1.2D + 1.6WL7+0.5Lr	7.7	0.70	0.00	3.41	1.25	0.00	6.66	4.66	2.16	2.16	4.50
1.2D + 1.6WL8+0.5Lr	7.3	0.54	0.00	3.26	0.96	0.00	6.23	4.23	2.30	2.30	4.07
0.9D+1.6WL1	11.0	4.66	0.00	4.91	8.28	0.00	15.19	13.19	-3.37	-3.37	13.03
0.9D+1.6WL2	8.7	3.54	0.00	3.87	6.29	0.00	12.16	10.16	-2.42	-2.42	10.00
0.9D+1.6WL3	11.2	0.08	0.00	4.96	0.14	0.00	7.10	5.10	4.82	4.82	4.94
0.9D+1.6WL4	8.9	1.20	0.00	3.95	2.13	0.00	8.08	6.08	1.82	1.82	5.92
0.9D+1.6WL5	5.9	0.27	0.00	2.62	0.48	0.00	5.10	3.10	2.14	2.14	2.94
0.9D+1.6WL6	5.6	0.43	0.00	2.48	0.76	0.00	5.24	3.24	1.71	1.71	3.08
0.9D+1.6WL7	5.9	0.27	0.00	2.62	0.48	0.00	5.10	3.10	2.14	2.14	2.94
0.9D+1.6WL8	5.6	0.43	0.00	2.48	0.76	0.00	5.24	3.24	1.71	1.71	3.08

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	31-07-2020	31-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	31-07-2020	31-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

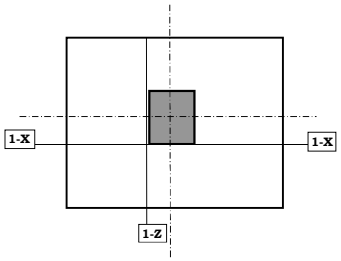
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2
= 3.004 m

Punching Area= ((0.55+2*0.5*0.201)*(0.55+2*0.5*0.201))
= 0.56 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)
= 2.91 T/m^2

Area of footing, A_f= 2.25 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 26.28 T

Strength of Concrete in shear Vc1 = 129.14 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 139.21 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 119.07 T **Table 22.6.5.2_ (c)**

Design Vc= 119.07 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (26.28*10000)/(3004*201)
= 0.44 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear ($0.75*\sqrt{fc}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	9.32	1.05	2.40	1.05	2.55	15.75	0.26	OK	0.74	0.127	OK	0.26
1.4D+1.7(MR)+0.5Lr	13.18	1.49	3.40	1.49	3.61	22.27	0.37	OK	0.74	0.18	OK	0.37
1.2D+1.6Lr+1.0(MR)	13.59	1.53	3.51	1.53	3.72	22.97	0.38	OK	0.74	0.185	OK	0.38
1.2D + 1.6WL1+0.5Lr	15.55	1.75	4.01	1.75	4.26	26.28	0.44	OK	0.74	0.212	OK	0.43
1.2D + 1.6WL2+0.5Lr	12.52	1.41	3.23	1.41	3.43	21.16	0.35	OK	0.74	0.171	OK	0.35
1.2D + 1.6WL3+0.5Lr	7.18	0.81	1.85	0.81	1.97	12.13	0.2	OK	0.74	0.098	OK	0.20
1.2D + 1.6WL4+0.5Lr	4.98	0.56	1.28	0.56	1.36	8.42	0.14	OK	0.74	0.068	OK	0.14
1.2D + 1.6WL5+0.5Lr	4.50	0.51	1.16	0.51	1.23	7.61	0.13	OK	0.74	0.061	OK	0.13
1.2D + 1.6WL6+0.5Lr	4.07	0.46	1.05	0.46	1.12	6.88	0.11	OK	0.74	0.055	OK	0.11
1.2D + 1.6WL7+0.5Lr	4.50	0.51	1.16	0.51	1.23	7.61	0.13	OK	0.74	0.061	OK	0.13
1.2D + 1.6WL8+0.5Lr	4.07	0.46	1.05	0.46	1.12	6.88	0.11	OK	0.74	0.055	OK	0.11
0.9D+1.6WL1	13.03	1.47	3.36	1.47	3.57	22.02	0.36	OK	0.74	0.178	OK	0.36
0.9D+1.6WL2	10.00	1.13	2.58	1.13	2.74	16.90	0.28	OK	0.74	0.136	OK	0.28
0.9D+1.6WL3	4.94	0.56	1.27	0.56	1.35	8.35	0.14	OK	0.74	0.067	OK	0.14
0.9D+1.6WL4	5.92	0.67	1.53	0.67	1.62	10.00	0.17	OK	0.74	0.081	OK	0.17
0.9D+1.6WL5	2.94	0.33	0.76	0.33	0.81	4.97	0.08	OK	0.74	0.04	OK	0.08
0.9D+1.6WL6	3.08	0.35	0.79	0.35	0.84	5.21	0.09	OK	0.74	0.042	OK	0.09
0.9D+1.6WL7	2.94	0.33	0.76	0.33	0.81	4.97	0.08	OK	0.74	0.04	OK	0.08
0.9D+1.6WL8	3.08	0.35	0.79	0.35	0.84	5.21	0.09	OK	0.74	0.042	OK	0.09

Load Case	kn (ϕ=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.35	0.001	217
1.4D+1.7(MR)+0.5Lr	0.49	0.001	217
1.2D+1.6Lr+1.0(MR)	0.51	0.001	217
1.2D + 1.6WL1+0.5Lr	0.58	0.001	217
1.2D + 1.6WL2+0.5Lr	0.47	0.001	217
1.2D + 1.6WL3+0.5Lr	0.27	0.001	217
1.2D + 1.6WL4+0.5Lr	0.19	0	0
1.2D + 1.6WL5+0.5Lr	0.17	0	0
1.2D + 1.6WL6+0.5Lr	0.15	0	0
1.2D + 1.6WL7+0.5Lr	0.17	0	0
1.2D + 1.6WL8+0.5Lr	0.15	0	0
0.9D+1.6WL1	0.49	0.001	217
0.9D+1.6WL2	0.37	0.001	217
0.9D+1.6WL3	0.18	0	0
0.9D+1.6WL4	0.22	0.001	217
0.9D+1.6WL5	0.11	0	0
0.9D+1.6WL6	0.11	0	0
0.9D+1.6WL7	0.11	0	0
0.9D+1.6WL8	0.11	0	0

Location	Mux (T-m)	ϕ Kn = M_{ul}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd. (mm)
At bottom	1.75	0.43	0.58	0.001	217.00	390.60	12ϕT	289.55	300	200
At top	0.00	0.00	0.00	0.0018	361.80	361.80	12ϕT	312.60	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

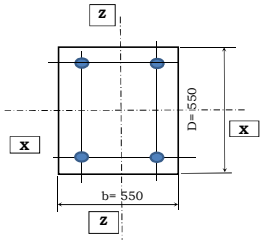
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2	3
Date	31-07-2020	31-08-2020	26-10-2023	Date	31-07-2020	31-08-2020	26-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT	
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI	

DESIGN OF COLUMN MKD P1

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1700

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) = 1700.00
Unsupported length of column about Z (l_{uz}) = 1700.00

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 21.42 < 22 SECTION 6.2.5a — Radius of gyration r_x= 159

$K\,l_{uz} / r_z$ = 21.42 < 22 SECTION 6.2.5a — Radius of gyration r_z= 159

γ_y =	0.68
γ_z =	0.68

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	1.36	4.49	0.00	3.6	6.9	0.0	0.01
1.4D+1.7(MR)+0.5Lr	1.55	10.40	0.00	4.1	16.0	0.0	0.02
1.2D+1.6Lr+1.0(MR)	1.79	9.44	0.00	4.7	14.5	0.0	0.01
1.2D + 1.6WL1+0.5Lr	2.82	5.64	0.00	7.4	8.7	0.0	0.01
1.2D + 1.6WL2+0.5Lr	2.26	3.30	0.00	5.9	5.1	0.0	0.00
1.2D + 1.6WL3+0.5Lr	0.45	5.75	0.00	1.2	8.8	0.0	0.01
1.2D + 1.6WL4+0.5Lr	-0.11	3.48	0.00	0.3	5.4	0.0	0.01
1.2D + 1.6WL5+0.5Lr	0.35	0.49	0.00	0.9	0.7	0.0	0.00
1.2D + 1.6WL6+0.5Lr	0.27	0.17	0.00	0.7	0.3	0.0	0.00
1.2D + 1.6WL7+0.5Lr	0.35	0.49	0.00	0.9	0.7	0.0	0.00
1.2D + 1.6WL8+0.5Lr	0.27	0.17	0.00	0.7	0.3	0.0	0.00
0.9D+1.6WL1	2.33	3.87	0.00	6.1	5.9	0.0	0.01
0.9D+1.6WL2	1.77	1.53	0.00	4.6	2.4	0.0	0.00
0.9D+1.6WL3	-0.04	3.98	0.00	0.1	6.1	0.0	0.01
0.9D+1.6WL4	-0.60	1.71	0.00	1.6	2.6	0.0	0.00
0.9D+1.6WL5	-0.14	-1.29	0.00	0.4	-2.0	0.0	0.00
0.9D+1.6WL6	-0.22	-1.61	0.00	0.6	-2.5	0.0	0.00
0.9D+1.6WL7	-0.14	-1.29	0.00	0.4	-2.0	0.0	0.00
0.9D+1.6WL8	-0.22	-1.61	0.00	0.6	-2.5	0.0	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. =	3221 Sq mm
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)		
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)	Total A_{sc} =	1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1025.65375 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cbD^2$						
1.4D	0.05	0.05	29.1	29.12	0.55	1.16	0.09	O.K
1.4D+1.7(MR)+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.10	O.K
1.2D+1.6Lr+1.0(MR)	0.05	0.05	29.1	29.12	0.55	1.16	0.12	O.K
1.2D + 1.6WL1+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.20	O.K
1.2D + 1.6WL2+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.16	O.K
1.2D + 1.6WL3+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.02	O.K
1.2D + 1.6WL4+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.00	O.K
1.2D + 1.6WL5+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.02	O.K
1.2D + 1.6WL6+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.01	O.K
1.2D + 1.6WL7+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.02	O.K
1.2D + 1.6WL8+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.01	O.K
0.9D+1.6WL1	0.05	0.05	29.1	29.12	0.55	1.16	0.16	O.K
0.9D+1.6WL2	0.05	0.05	29.1	29.12	0.55	1.16	0.12	O.K
0.9D+1.6WL3	0.05	0.05	29.1	29.12	0.55	1.16	0.00	O.K
0.9D+1.6WL4	0.05	0.05	29.1	29.12	0.55	1.16	0.03	O.K
0.9D+1.6WL5	0.05	0.05	29.1	29.12	0.55	1.16	0.01	O.K
0.9D+1.6WL6	0.05	0.05	29.1	29.12	0.55	1.16	0.01	O.K
0.9D+1.6WL7	0.05	0.05	29.1	29.12	0.55	1.16	0.01	O.K
0.9D+1.6WL8	0.05	0.05	29.1	29.12	0.55	1.16	0.01	O.K

Links, **10mm dia @ 200 C/C**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	31-07-2020	31-08-2020	26-10-2023	Date	31-07-2020	31-08-2020	D01
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS 001	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS 001	26-10-2023
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] PUMP SHED DETAIL DRAWING NO. MD-216-49PR-EG-ST-GAD-2424
[7] PUMP SHED STRUCTURAL DOC NO. MD-216-49PR-EG-ST-CAL-2425

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	31-07-2020	31-08-2020	26-10-2023				Date	31-07-2020	31-08-2020	26-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 Foundation Load Data for Pump Shed Columns

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	31-07-2020	31-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	31-07-2020	31-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 MARKED AS C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

C01 MMGP:POWER PLANT: Pump Shed Load Data (DATED 27.07.2020)

1 DL	Dead load	
2 LLROOF	Live load on roof	
21 WL1(+X)CASE-A	Wind load	Wind load in direction of +X Suction(from A grid to D grid)
22 WL2(+X)CASE-B	Wind load	Wind load in direction of +X Pressure(from A grid to D grid)
23 WL3(-X)CASE-A	Wind load	Wind load in direction of -X Suction(from D grid to A grid)
24 WL4(-X)CASE-B	Wind load	Wind load in direction of -X Pressure(from D grid to A grid)
25 WL5(+Z)CASE-A	Wind load	Wind load in direction of +Z Suction(from line-1 to line-3)
26 WL6(+Z)CASE-B	Wind load	Wind load in direction of +Z Pressure(from line-1 to line-3)
27 WL7(-Z)CASE-A	Wind load	Wind load in direction of -Z Suction(from line-3 to line-1)
28 WL8(-Z)CASE-B	Wind load	Wind load in direction of -Z Pressure(from line-3 to line-1)
30 MR	Monorail load	

Case-A and Case-B defined as per ASCE-7-16 fig-27.3-5

Monorail and wind load cases are exclusive.

Sign convention

FY	+ve sign indicates vertical load downward
	-ve sign indicates vertical load upnward
FX	+ve sign indicates horizonal load in across direction from grid-A to D
	-ve sign indicates horizonal load in across direction from grid-D to A
FZ	+ve sign indicates horizonal load in along direction from line 1 to 3
	-ve sign indicates horizonal load in along direction from line 3 to 1

LOAD DATA TABLE

Column Grid	Column Grid	Load cases	Horizontal Fx Ton	Vertical Fy Ton	Horizontal Fz Ton
A	3	1 DL	0.14	1.17	-0.18
		2 LLROOF	0.2	0.81	-0.21
		21 WL1(+X)CASE-A	0.8	0.6	-0.12
		22 WL2(+X)CASE-B	0.63	-0.11	0.07
		23 WL3(-X)CASE-A	-0.63	0.06	-0.05
		24 WL4(-X)CASE-B	-0.84	-0.94	0.16
		25 WL5(+Z)CASE-A	-0.32	-1.98	0.9
		26 WL6(+Z)CASE-B	-0.36	-2.18	0.94
		27 WL7(-Z)CASE-A	-0.32	-0.63	-0.24
		28 WL8(-Z)CASE-B	-0.34	-0.73	-0.21
		30 MR LOAD	1.23	3	0
A	1	1 DL	-0.14	1.17	-0.18
		2 LLROOF	-0.2	0.81	-0.21
		21 WL1(+X)CASE-A	0.63	0.11	-0.07
		22 WL2(+X)CASE-B	0.8	-0.6	0.12
		23 WL3(-X)CASE-A	-0.77	0.55	-0.1
		24 WL4(-X)CASE-B	-0.56	-0.23	0.08
		25 WL5(+Z)CASE-A	0.32	-1.98	0.9
		26 WL6(+Z)CASE-B	0.36	-2.18	0.94
		27 WL7(-Z)CASE-A	0.32	-0.63	-0.24
		28 WL8(-Z)CASE-B	0.34	-0.73	-0.21
		30 MR LOAD	-1.23	3	0
B	3	1 DL	0.97	3.21	0
		2 LLROOF	0.39	1.62	0
		21 WL1(+X)CASE-A	0.91	0.61	0
		22 WL2(+X)CASE-B	0.56	-0.85	0
		23 WL3(-X)CASE-A	-0.57	0.68	0
		24 WL4(-X)CASE-B	-0.92	-0.74	0
		25 WL5(+Z)CASE-A	-0.63	-2.61	0
		26 WL6(+Z)CASE-B	-0.68	-2.81	0
		27 WL7(-Z)CASE-A	-0.63	-2.61	0
		28 WL8(-Z)CASE-B	-0.68	-2.81	0
		30 MR LOAD	0	3	0
B	1	1 DL	-0.97	3.21	0
		2 LLROOF	-0.39	1.62	0
		21 WL1(+X)CASE-A	0.56	0.85	0
		22 WL2(+X)CASE-B	0.91	-0.61	0
		23 WL3(-X)CASE-A	-0.86	0.53	0
		24 WL4(-X)CASE-B	-0.52	-0.92	0
		25 WL5(+Z)CASE-A	0.63	-2.61	0
		26 WL6(+Z)CASE-B	0.68	-2.81	0
		27 WL7(-Z)CASE-A	0.63	-2.61	0
		28 WL8(-Z)CASE-B	0.68	-2.81	0
		30 MR LOAD	0	3	0
other columns	3	1 DL	0.19	1.43	0.18
		2 LLROOF	0.39	1.62	0.21
		21 WL1(+X)CASE-A	0.85	0.46	0.06
		22 WL2(+X)CASE-B	0.5	-1	-0.13
		23 WL3(-X)CASE-A	-0.52	0.81	0.1
		24 WL4(-X)CASE-B	-0.84	-0.56	-0.09
		25 WL5(+Z)CASE-A	-0.63	-1.94	-0.05
		26 WL6(+Z)CASE-B	-0.68	-2.14	-0.08
		27 WL7(-Z)CASE-A	-0.63	-3.29	-0.62
		28 WL8(-Z)CASE-B	-0.68	-3.49	-0.65
		30 MR LOAD	-1.23	3	0
other columns	1	1 DL	-0.19	1.43	0.18
		2 LLROOF	-0.39	1.62	0.21
		21 WL1(+X)CASE-A	0.5	1	0.13
		22 WL2(+X)CASE-B	0.85	-0.46	-0.06
		23 WL3(-X)CASE-A	-0.81	0.39	0.05
		24 WL4(-X)CASE-B	-0.45	-1.1	-0.15
		25 WL5(+Z)CASE-A	0.63	-1.94	-0.05
		26 WL6(+Z)CASE-B	0.68	-2.14	-0.08
		27 WL7(-Z)CASE-A	0.63	-3.29	-0.62
		28 WL8(-Z)CASE-B	0.68	-3.49	-0.65
		30 MR LOAD	1.23	3	0

(Refer drg no. MD-216-49PR-EG-ST-GAD-2424 for other details)

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	31-07-2020	31-08-2020	26-10-2023			Date		31-07-2020	31-08-2020	26-10-2023		
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Prepared By	SHD	SHD	SHD			Prepared By		SHD	SHD	SHD		
Checked	SGC	SGC	SGC			Checked		SGC	SGC	SGC		
Approved	PRI	PRI	PRI			Approved		PRI	PRI	PRI		

C01

ATTACHMENT 2 Comments Resolution Sheet

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT**	**PROJECT NO.: 216**	
STAGE:	**4 - PROJECT EXECUTION.**	**REVIEW LOG DOC. NO. & REV.:**	
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-004367, 20 August 2020		

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 20 August 2020	Comment by	EPC Response DATE: 31 August 2020	Status	Remarks
			General Comments (cc)				
MD-216-49PR-EG-CV-CAL-2016-A01	49PR	1	Document to be updated with final Load Data of Pump Shed (Pg no. 13)	Mr. Doron Golan	Noted & Updated.	Closed	